Software Engineering II
Semester 2026-I
Workshop No. 2 — Design Artifacts and System Modeling

Eng. Liliana Marcela Olarte, M.Sc.
Eng. Carlos Andrés Sierra, M.Sc.
Lecturers
Computer Engineering Program
School of Engineering
Universidad Nacional de Colombia

Welcome to Workshop 2! This session focuses on the *design phase* of your *Software Engineering II* course project. You will produce key design artifacts and models to clarify your system's structure and prepare for implementation.

Scope and Objectives

- **CRC Cards:** Define the main classes, their responsibilities, and collaborators using CRC cards.
- **Mockups:** Create mockups or wireframes for the main screens of your application.
- **Business Model Processes:** Document essential business processes using activity diagrams or BPMN.
- **Architecture Diagram:** Draw the overall software architecture, showing main components and their interactions.
- **Class Diagram:** Create UML class diagrams for your system, including attributes, methods, and relationships.
- **Relational Database Model:** Design the relational database schema, including tables, keys, and relationships.

Carlos Andrés Sierra, Computer Engineer, M.Sc. in Computer Engineering, Lecturer at Universidad Nacional de Colombia.
Any comment or concern regarding this workshop can be sent to Carlos A. Sierra at: *casierrav@unal.edu.co*.

Methodology and Deliverables

1. **CRC Cards**
    - Identify and describe the main classes in your system.
    - For each class, specify its responsibilities and collaborators.
    - Present CRC cards in a table or diagram format (one per each class).
2. **Mockups**
    - Create mockups or wireframes for your screens.
    - You may use digital tools (e.g., Figma, Balsamiq) or hand-drawn sketches.
    - Briefly explain the purpose of each screen.
3. **Business Model Processes**
    - Document at least one core business process using an activity diagram or BPMN.
    - Describe the process and its role in your application.
    - Reference: `https://www.visual-paradigm.com/guide/bpmn/what-is-bpmn/`
4. **Architecture Diagram**
    - Draw a software architecture diagram showing system components and their interactions.
    - Indicate technologies, layers, and communication flows.
    - Reference: `https://www.lucidchart.com/blog/how-to-draw-architectural-diagrams`
5. **Class Diagram**
    - Create UML class diagrams for the main classes.
    - Include attributes, methods, and relationships.
    - Reference: `https://www.uml-diagrams.org/class-diagrams-overview.html`
6. **Relational Database Model**
    - Design the relational database schema for your application.
    - Include tables, primary and foreign keys, and relationships.
    - Present the model as an ER diagram or table definitions.
7. **Delivery Format**
    - Compile all deliverables into a single PDF.
    - Organize your files in a folder named `Workshop-2` in your course project repository, with a `README.md` referencing each section.

Project Requirements Checklist

- CRC cards for main classes.
- Mockups for key screens.
- Business process documentation.
- Architecture and class diagrams.
- Relational database model.
- Organized and referenced documentation.

Deadline

Friday, July 17th, 2026, at 22:00. Late submissions will not be accepted.

Notes

- All documents must be in **English**.
- Cite any references (articles, tutorials, tools) that influenced your design choices.
- Focus on clarity and completeness. This design phase will guide your implementation in future workshops.

Good luck! A clear and well-documented design will set the stage for a successful project implementation.